ASSIGNMENT 10 – XV6 OPERATING SYSTEM

Implementation and Testing Report

1. Introduction

This assignment extends the xv6 teaching operating system with memory-management inspection facilities and a user-level simulation of page-replacement algorithms. The work is organized into four parts: physical-memory statistics, virtual-to-physical address translation, page-table inspection and protection, and comparison of FIFO, Least Recently Used (LRU), and Optimal page replacement.

2. Implementation

Q1 – Physical memory statistics

A system call named memstats() was added to expose physical-page allocation statistics. The implementation tracks pages allocated and freed by the kernel allocator and reports the current allocated-page count and available free pages. The user test allocates five pages, checks the statistics, releases the pages, and checks that the counts return to their earlier level.

Observed test values: before allocation, 425 pages had been allocated and 214 freed (211 currently allocated, 56,790 available). After allocating five pages, the totals became 430 allocated, 214 freed (216 currently allocated, 56,785 available). After freeing them, the totals became 430 allocated, 219 freed (211 currently allocated, 56,790 available).

Q2 – Virtual-to-physical address translation

A vtop() kernel routine and corresponding system-call interface were added to translate a user virtual address to its physical address using the process page table. The routine reports 0xffffffff for an unmapped address. The test program prints the virtual and physical addresses of a variable. The observed addresses were 0x2FCC and 0x0DEDFFCC; their page offsets match (0xFCC), as expected.

Q3 – Page-table inspection and read-only protection

Two system-call operations were added. pageinfo() reports the virtual address, physical address, page-table entry, and the present, writable, and user-accessible flags. setreadonly() clears the writable bit in the corresponding page-table entry and reloads the page-directory base register to flush the TLB. The test showed the entry changing from 0xdfbc067 to 0xdfbc065: the present and user bits remained set, while the writable bit changed from 1 to 0.

Q4 – Page-replacement algorithms

A user program, pagereplace, implements FIFO, LRU, and Optimal replacement for the reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. For each reference with three frames, the program displays the frame contents and whether the reference is a hit or fault. It then calculates faults, hits, hit ratio, and fault ratio for three, four, and five frames. Ratios are displayed as fractions out of 20 because xv6 printf does not support floating-point format specifiers.

3. Page-replacement results

Frames	Algorithm	Faults	Hits	Hit ratio	Fault ratio
3	FIFO	15	5	5/20	15/20
3	LRU	12	8	8/20	12/20
3	Optimal	9	11	11/20	9/20
4	FIFO	10	10	10/20	10/20
4	LRU	8	12	12/20	8/20
4	Optimal	8	12	12/20	8/20
5	FIFO	9	11	11/20	9/20
5	LRU	7	13	13/20	7/20
5	Optimal	7	13	13/20	7/20

The results show that Optimal produces the fewest faults for three frames. With four and five frames, Optimal and LRU have the same fault counts for this reference string. Increasing the number of frames reduces or preserves the number of faults in these observed runs.

4. Testing and submission contents

The four user test programs (memtest.c, vttest.c, pagetest.c, and pagereplace.c) were compiled into xv6 and executed from the xv6 shell. The observed outputs confirmed the expected memory-statistic changes, address translation, page-table permission change, and page-replacement counts. The submission is organized with question-wise folders containing modified source files, diff files, and screenshots, alongside a common modified_files folder containing the union of modified files (latest version retained for repeated filenames) and a test folder containing the four test programs.

5. Conclusion

The assignment successfully adds kernel interfaces for examining physical memory and process page tables, demonstrates virtual-to-physical translation and read-only page protection, and compares three standard page-replacement policies. The test results are consistent with the expected behavior for the supplied reference string.